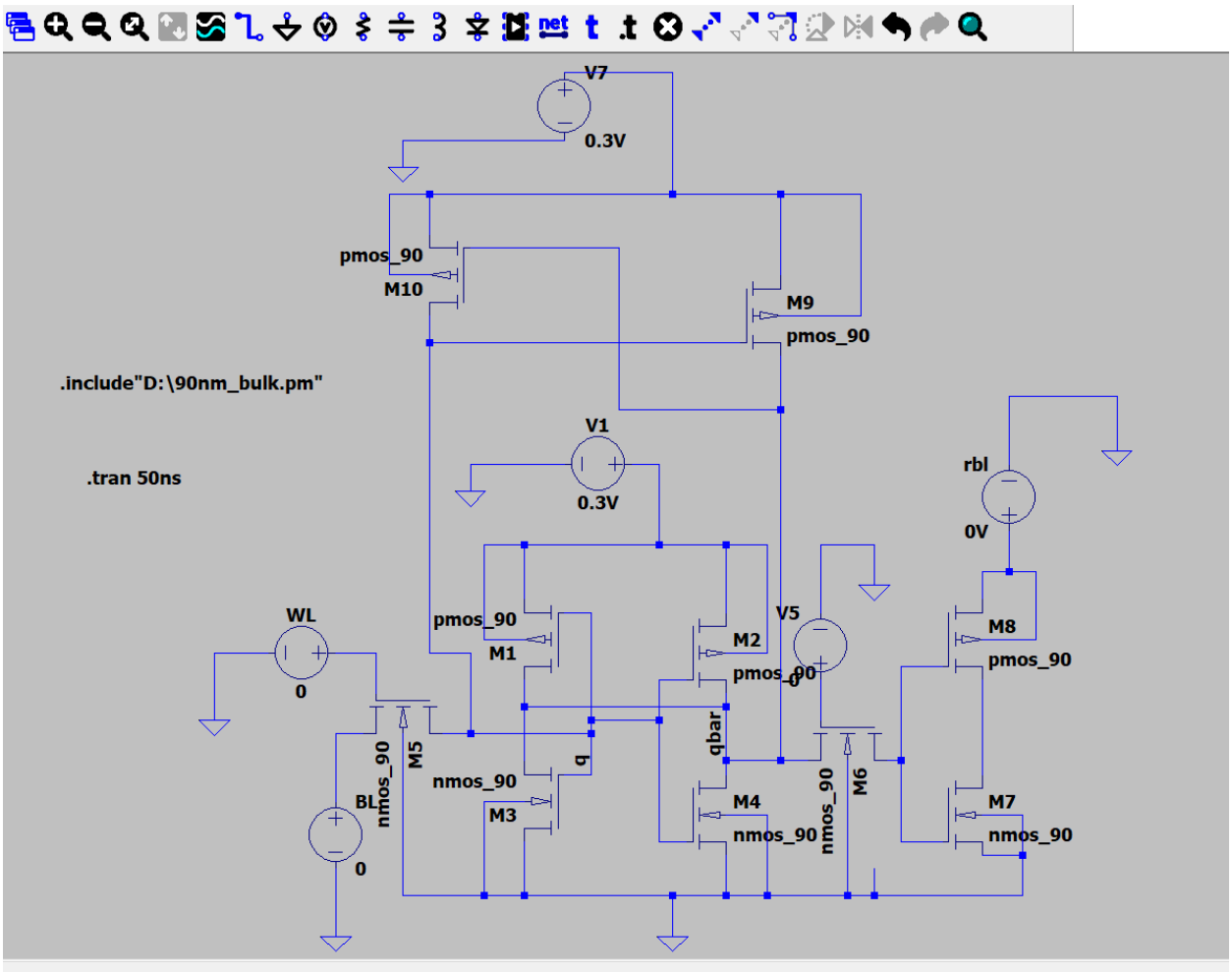

Fig 01 : proposed circuit (read and write port differen Having separate read and write ports increases both the write and read SNM. ,add two pmos header part this part reduce power)

1. https://www.atlantis-press.com/proceedings/ramsita-25/126011533

(This paper using 90 nm cmos technology at Cadence virtuoso tool)

(This paper presents an analysis of 6T, 7T, and 8T SRAM cells. The study evaluates the write SNM (WSNM) and read SNM (RSNM) at supply voltages of 0.3 V and 1 V. At 0.3 V, the read and write power, read and write delays, and leakage power are carefully measured. The results demonstrate that these SRAM designs are suitable for low-power applications, providing both stability and energy efficiency.

	Paper 1	Proposed circuit
Technology and tool	90 nm cmos technology Cadence virtuoso tool	90 nm cmos technology and Ltspice
Literature Review	This paper presents an analysis of 6T, 7T, and 8T	Compromise with this paper and determine which

	SRAM cells. The study evaluates the write SNM (WSNM) and read SNM (RSNM) at supply voltages of 0.3 V and 1 V. At 0.3 V, the read and write power, read and write delays, and leakage power are carefully measured. The results demonstrate that these SRAM designs are suitable for low-power applications, providing both stability and energy efficiency.	parameters have shown improvement.
Read power Consumption of 6T,7T and 8T at 300mv	1.6nw,2.7nw,2.7nw	363.88 pW(has improved)
Read delay of 6T,7T and 8T at 300mv	23.2pS,81.1ps,54.4pS	delay 376 ps (has not improved)
Write power Consumption of 6T,7Tand 8T at 300mv	5.6nW,3nW,6.1nW	2.83 nW(has improved)
Write delay of 6T,7T and 8T at 300mv	482pS,1.5ns,482ns	743ps(has not improved)
Leakage power of 6T,7T and 8T	1.5nW,0.7nW,2.3nW	1.51 nW (which is better compared to the 8T SRAM)
RSNM of 6T,7T and 8T(at 1V)	42mv,580mv,278 mv	482.35 mV (improved from 6T and 8T)
RSNM of 6T,7T and 8T(at 300mv)	13mv,133mv,78mv	146.4 mV(improved)
WSNM of 6T,7T and 8T(at 1V)	421mv,270mv,421mv	476.5 mv(improved)
WSNM of 6T,7T and 8T(at 300mV)	132mv,71mv,132mv	144.6 mv

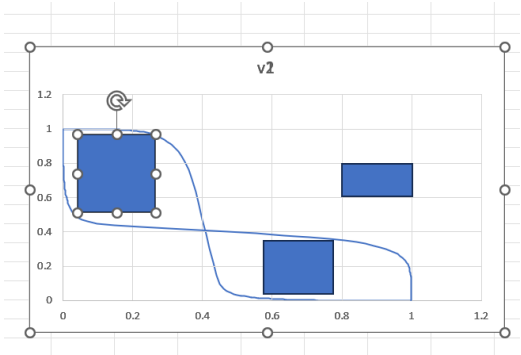

Fig 02 : RSNM proposed circuit 482.35 mv (in 1 V)

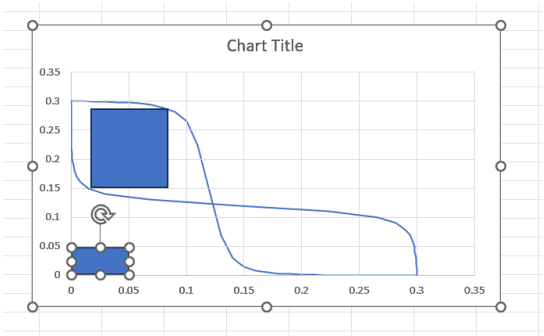

Fig 03 : RSNM of proposed circuit 146.4mv(at 300mv)

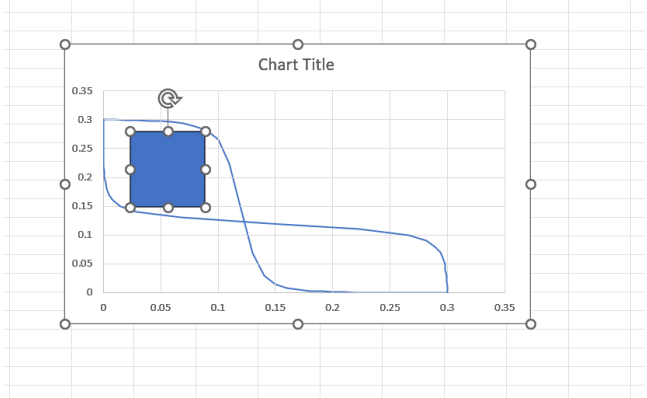

Fig 04 : WSNM of proposed circuit 144.6 mv (at 300mv)

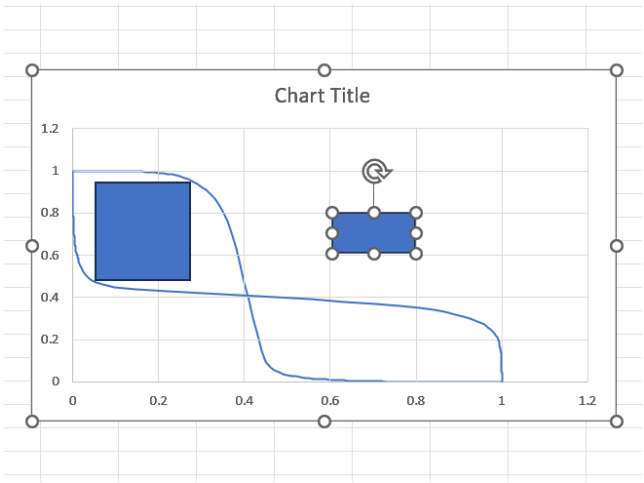

Fig 05 : WSNM of proposed circuit 476.5 mv (at 1V)

2. https://link.springer.com/chapter/10.1007/978-981-13-3393-4_47

(this paper using 45 nm cmos technology in cadence virtuoso)

(This paper proposed 5 sram circuit and compromise with basic 6T sram cell)

	Paper 2	Proposed circuit
Technology and tool	45 nm cmos technology in cadence virtuoso	90 nm cmos technology and Ltspice
Literature Review	(This paper proposed 5 sram circuit and compromise with basic 6T sram cell)	Compromise with this paper and determine which parameters have shown improvement.
RSNM of 6T,circuit(1,2,3,4 ,5)	180mv,40mv,96mv,20mv,35mv,150mv(1.1V)	482.35 mV (hasimproved) at 1V
WSNM of 6T,circuit(1,2,3,4 ,5)	360mv,300mv,425mv,380mv,250mv,200mv (at 1.1V)	476.5 mv(has improved) at 1V
Leakage power of 6T,circuit(1,2,3,4 ,5)	526nW,29.18nW,22.3nW,1.9nW,0.07nW,20.4nW	1.51nW (improved but has not improved from circuit 4)
Total power of 6T,circuit(1,2,3,4 ,5)	8.8nW,83.21nW,3.2nW,174nW,2.08nW,5.6nW	363.88pW(read power),2.83nW(write power) improved but has not improved from circuit 4

3. https://link.springer.com/chapter/10.1007/978-981-16-3767-4_19

(this paper use 45 nm cmos technology at cadence virtuoso)

This paper compromise 10 T sram ,8T sram and 6 T sram.

	Paper3	Proposed circuit
Technology and tool	45 nm cmos technology at cadence virtuoso	90 nm cmos technology and Ltspice
Read power	10 T sram read power approximately 100nW(at 0.5V)	363.88 pW at 300mv (improved)
Write power	10 T sram write power approximately 50nW(at 0.5V	2.83 nW at 300 mv (improved)
Read delay	10T sram read delay 400 pS at 0.5V	376 pS read delay at 300 mv
Write delay	10T sram write 2.25 nS at 0.5V	743ps at 300mv
RSNM	10T sram RSNM 375 mv at 1V	RSNM 482.35 at 1V (improved)
WSNM	10T sram WSNM 450 mv at 0.9V	WSNM 476.5 mv at 1V (improved)
Leakage power	10 T sram Leakage power 20pW at 1V	1.51 nW at 300mv (has not improved)

4. https://link.springer.com/chapter/10.1007/978-981-16-3728-5_4

(32 nm cmos Technology at HSPICE tool)

0.9 V(operating voltage vdd = 0.9V)

This paper compared to proposed 9T sram to 6 T sram.

	Paper 4	Proposed circuit
Technology and tool	32 nm cmos Technology at HSPICE tool	90 nm cmos technology and Ltspice
Literature Review	compared to proposed 9T sram to 6 T sram	Compromise with this paper and determine which parameters have shown improvement
RSNM of proposed 9T sram and 6T sram at 0.9V	244mv,57mv	482.35 mV at 1V (improved)
WSNM of proposed 9T sram and 6T sram at 0.9V	405mv,383mv	WSNM 476.5 mv at 1V (improved)
Leakage power of proposed 9T sram and 6T sram at 0.9V	3.66nW,7.51nW	1.51nW at 1V (improved)

5. https://ietresearch.onlinelibrary.wiley.com/doi/abs/10.1049/iet-cdt.2019.0234

(using UMC 65-nm process technology)

This Paper compare to proposed 11T sram to PG9T ,9T,10T,6T sram

(vdd = 1.2V)

	Paper 5	Proposed circuit
Technology and tool	using UMC 65-nm process technology	90 nm cmos technology and Ltspice
Literature Review	This Paper compare to proposed 11T sram to PG9T ,9T,10T,6T sram	Compromise with this paper and determine which parameters have shown improvement
RSNM	Proposed 11 T sram RSNM 460mv at 1.2V	482.35 mV at 1V
WSNM	Proposed 11 T sram RSNM 510 mv at 1.2V	WSNM 476.5 mv at 1V
Read power	Proposed 11 T sram Read power 18000nW at 1.2V	363.88 pW at 300mv (improved)
Write power	Proposed 11 T sram write power 18000nW at 1.2V	2.83 nW at 300 mv (improved)
Read delay	4.8ns at FF process corner(nmos fast pmos fast)	delay 376 ps (improved)
Write delay	1.9ns at FF process corner(nmos fast pmos fast)	743ps(improved)
Leakage power	Proposed 11 T sram write power 5000nW at 0.3V	1.51nW at 300mv (improved)

6. https://www.researchsquare.com/article/rs-7923985/latest

(using 90nm cmos technology in cadence virtuoso)

This paper compromise improved Stacked Based 10T SRAM andMTCMOS-Based 10T SRAM with conventional 6T,8T,10T,12T sram.

(operating voltage vdd = 1.8V)

	Paper 6	Proposed circuit
Technology and tool	90nm cmos technology in cadence virtuoso	90 nm cmos technology and Ltspice
Literature Review	This paper compromise improved Stacked Based 10T SRAM andMTCMOS-Based 10T SRAM with conventional 6T,8T,10T,12T sram	Compromise with this paper and determine which parameters have shown improvement
SNM	Proposed 10T MTCOMS 379.456 mv at 1.8V	WSNM 476.5 mv at 1V RSNM 482.35 mV at 1V(improved)

Power consumption	Proposed 10T MTCOMS 13800nw mv at 1.8V	Read power363.88 pW at 300mv Write power 2.83 nW at 300 mv (improved)

7. https://ieeexplore.ieee.org/abstract/document/11172156/

(using 90nm cmos technology in cadence virtuoso)

This paper compromise 6T,8T,10T and 12T sram .
(operating voltage vdd = 1.8 V)

Snm and power consumption significantly improved from this paper.

I have found these points:

1. The proposed circuit is demonstrated to be applicable at lower voltages by evaluating the SNM at low voltage, as SNM indicates the stability of the circuit.

2. The reduced power consumption of the proposed circuit during read and write operations confirms its applicability as an ultra-low-power SRAM design.

3. A lower leakage power in the proposed circuit indicates its improved energy efficiency and confirms its suitability for low-power applications.